

University of Mosul
College of Computer Sciences and
Mathematics
Computer Science Dept. /First Class

Principles of Computer
Organization
2025-2026

Lec. 11

presentation
notes

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Microprocessor 8086/8088



8086 CPU organized as two separate processors called the Bus Interface Unit (BIU) and the Execution Unit (EU). The BIU provides hardware functions, including generation of the memory and I/O addresses for the transfer of the data between the outside of the CPU and the EU.

The EU receives program instruction code and data from the BIU, execute these instructions and store the result in the general registers. By passing the data back to the BIU, data can also be stored in a memory location or written to an output device. Note that the EU has no connection to the system buses. It receives and output all its data through the BIU.

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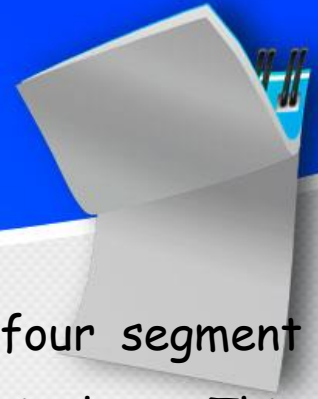


The only difference between an 8088 μP and 8086 μP is the BIU. In the 8088, the BIU data bus path is 8 bits wide versus the 8086's 16 bits data bus. Another difference is that the 8088 instruction queue is four bytes long instead of six.

From the previous figure we see that the main parts of the BIU are:

- *The instruction queue.*
- *A register arrays.*
- *An adder.*
- *A control unit.*

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The register array serves addressing purposes. It includes four segment registers and an instruction pointer (IP). All of which are 16 bit long. This particular adder is used only for address calculations. It does not pass the full arithmetic /logic capabilities of the main ALU of the EU. Three temporary registers shown with the adder hold either address components (prior to an address calculation) or the result. Notice that each unit of the 8088/8086 has its own independent control section. The control section of the BIU is responsible for all data transfers over the external CPU bus. Incoming data flow via the internal bus of the BIU to the instruction queue or its register address.

Addresses flow from the temporary register, at the output of the adder to the external CPU bus via a 20 bit address bus.

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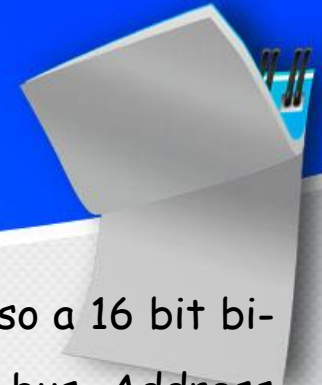


The main parts of the EU are:

- The array of general registers.
- The main ALU.
- A flag register.
- A control section.

There are again three temporary registers holding operands during data manipulation by the ALU. Employed in most CPUs, such registers are generally inaccessible and transparent to programmer. Note that all registers of the EU connect to a common 16 bit wide bus for data exchanges, results from the ALU flow to registers through this bus as well. The control section of the EU is more complex than that of the BIU. Some of the control lines of the external bus as, for example, bus control and interrupt lines, are handled by the control section of the EU. However, all data transfer are via the BIU. Opcode flows from the instruction queue into the instruction register (IR) for decoding

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Operands flow to appropriate registers via the ALU data bus. Notice also a 16 bit bi-directional data path between the BIU register array and the ALU data bus. Address components for final address calculation at the BIU as well as data destined for the external bus flow from the EU via this path.

Memory Segmentation:

When the EU of the 8088/8086 decodes an instruction, it determines (among the other things) the applicable addressing modes. Then it manipulates each operand field according to derive a 16 bit address. Since internal paths are 16 bit wide, the EU is generally limited to handling quantities up to 16 bits long. A 16 bit address allows unique identification of 64 Kilo byte (2^{16}) byte location. But since the 8088/8086 has a 1 Mega byte address space, the 16 bits address must be argumented somehow to 20 bits to make possible unique identification of 1 million byte locations (1,024,576).

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Here is exactly where the role of the BIU comes into play (in address calculation). The BIU takes the 16 bit address derived by the EU and adds it to the contents of a register. This register can be any one of the four so-called "segment registers". Before the addition, the contents of the segment register are shifted four positions in the direction of the MSB. That is, to the left. Thus, the adder produces a 20 bit quantity as in the figure. This quantity is used for identification of physical memory locations and is therefore called "physical address". Addresses presented over the external bus are always physical addresses. However, the address component it contributes is named a "segment address" rather than a base address. On the other hand, the address provided by the EU is treated as a displacement (or offset) what is the lowest physical address within the segment. Clearly, this corresponds to an all-zero offset and this coincides with the segment address (SA) itself.

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Advantages of segmented memory:

Segmented memory can be confusing at first. Simply, the program op-codes will be fetched from the code segment, while program data variables will be stored in the data and extra segments. Stack operations use register BP or SP and the stack segment.

An immediate advantage of having separate data and code segments is that one program work on several different sets of data. This is done by reloading register DS to point to the new data. Perhaps the greatest advantage of segmented memory is that programs that reference logical addresses only can be loaded and run anywhere in memory.

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This is because the logical addresses always range from 0000 to FFFFH, independent of the code segment base. Consider a multitasking environment in which the 8086/8088 is doing several different jobs at once. An inactive program can be temporarily saved on a magnetic disk and a new program, such programs are said to be relocatable, meaning that they will run at any location in memory. The requirements of writing relocatable programs are that no references be made to physical addresses, and no changes to the segment registers are allowed.

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Fetch and Execute:

Organization of the 8088/8086 CPU into a separate BIU and EU allows the fetch and execute cycles to overlap. To see this, consider what happens when the 8088/8086 is first started:

- 1.the BIU outputs the contents of the instruction pointer (IP) onto the address bus, cause the selected byte or word to be read into the BIU.*
- 2.Register IP is incremented by one to prepare for the next instruction fetch.*
- 3.Once inside the BIU, the instruction is passed to the "instruction queue". This is a First_In_First_Out (FIFO) storage (cache memory) linked to a pipeline.*

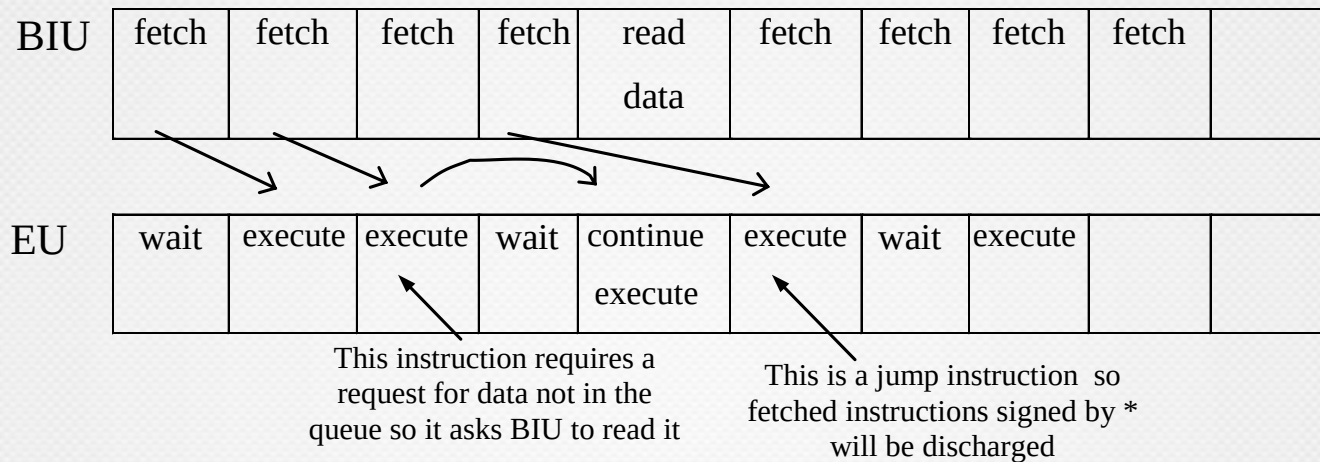
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1. Assuming that the queue is initially empty, the EU immediately draws this instruction from the queue and begins execution.
2. While the EU is executing this instruction, the BIU proceeds to fetch a new instruction, the BIU may fill the queue with several new instructions whenever the queue has room for one (with the 8088) or two (with 8086) additional bytes.

The advantage of this pipelined architecture is that the EU can execute instructions almost continually instead of having wait for the BIU to fetch a new instruction as in the following figure:

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There are three conditions that will cause the BIU and EU to enter a “wait” mode. The first occurs when an instruction requires access to a memory location not in the queue. The BIU must suspend fetching instructions and output the address of this memory location. After waiting for the memory access, the EU can resume executing instruction codes from the queue (and the BIU can resume filling the queue).

The second condition occurs when the instruction to be executed is a “Jump” instruction. In this case control is transferred to a new (non-sequential address). The queue, however, assumes that instruction will always be executed in sequence and this will be holding the “wrong” instruction codes.

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The EU must wait while the instruction at the jump address is fetched. Note that any bytes presently in the queue must be discarded (they are overwritten).

One another condition can cause the BIU to suspend fetching instructions. This occurs during execution of instructions that are slow to execute. For example, the instruction (AAM ASCII Adjust for Multiplication) requires 83 clock cycles to complete. At four clock cycles per instruction fetch, the queue will be completely filled during the execution of this single instruction. The BIU will thus have to wait for the EU to pull one or two bytes from the queue before resume the fetch cycle.

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Reading / writing data to 8086:

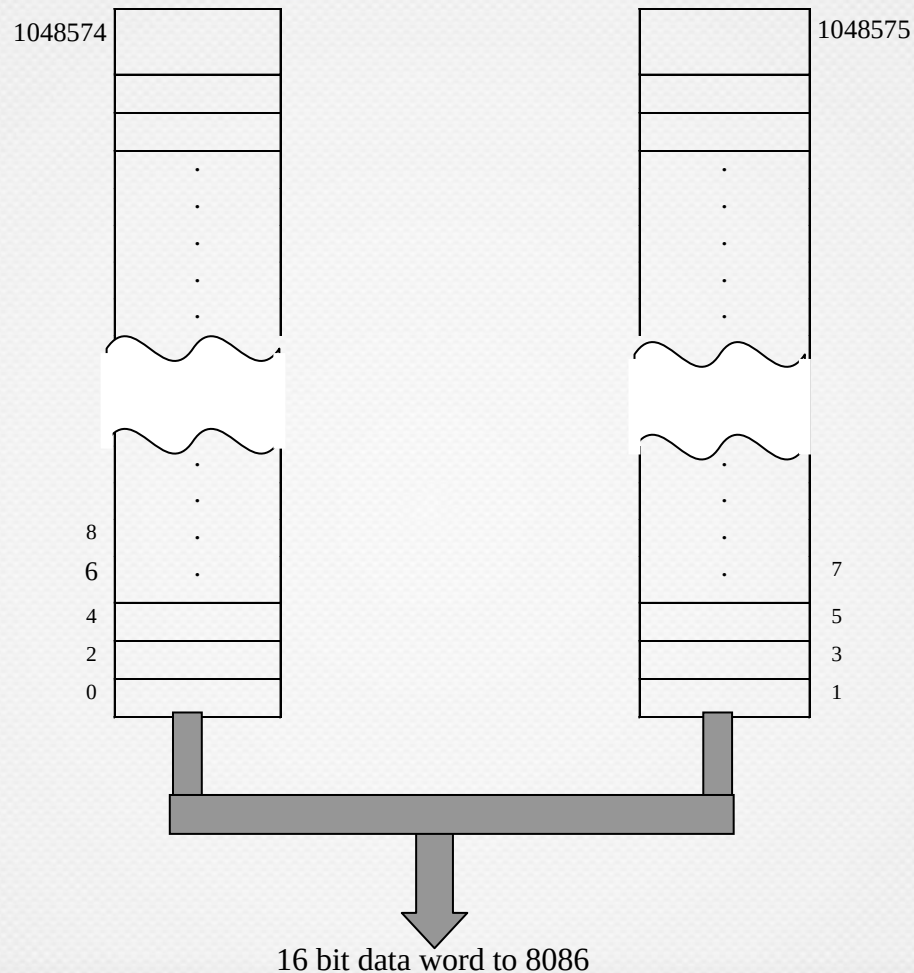
Even though it is considered a 16 bit MP (it has a 16 bit data bus width) its memory is still thought of in bytes. At first this might seem a disadvantage. Actually there are a couple of good reasons first it allows the processor to work on bytes as well as words. This is specially important with I/O devices such as printers, terminal, ..., etc (ASCII). Second many of the 8086s (and 8088s) operands are single bytes, other instructions may require anywhere from two to seven bytes. By being able to access individual bytes, these odd-lengthed instructions can be handled.

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The 8086 read 16 bits from memory by simultaneously addressed byte. For the reason the 8086 organized its memory into even-addressed bank and odd-addressed bank. 8086 provides control bus signals that can be decoded by the memory to determine if a byte or a word is to be accessed. If the CPU must access an odd word (e.g. the word stored in bytes 5 and 6), the CPU must perform two memory read cycles: one to fetch the low order byte and a second to fetch the high order byte. This slow down the processor but is transparent to the programmer.

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The 8088 with its 8-bit data bus interfaces to the 1MByte of memory as a single bank. Always when it is necessary to access a word (odd or even) two memory (read or write) cycles are performed, fortunately, for the programmer except for the slightly slower performance of the 8088, there is no difference between the two processors.